Dump-Resistor Energy Extraction for a High-Current Solid-State Switch

An alternative to repetitive MOV clamping for million-cycle test-bench duty

Davide Bagnara

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Abstract

A solid-state switch for a test bench must interrupt an inductive load ($R_L = 0.8\,\Omega$, $L = 2\,\mathrm{mH}$) fed from a 3 kV DC source at a peak current of 3600 A, with a target service life of several million operations. In the baseline design the inductive energy is absorbed by a metal-oxide varistor (MOV) bank clamping the switch at approximately 5.5 kV. This note shows that the MOV bank absorbs about 16.6 kJ per operation, a duty that is fundamentally incompatible with the repetitive-pulse capability of metal-oxide varistors, and develops an alternative in which the energy is extracted by a linear *dump resistor* switched across the load by a crowbar device. The circuit analysis, the energy budget, the commutation of the current into the dump branch, the sizing of the resistor and of the crowbar device, and the residual role of the MOV bank as a backup clamp are discussed quantitatively. The dump-resistor solution preserves an extinction speed comparable to the MOV clamp while moving essentially all of the cyclic energy into a component with practically unlimited electrical life, and, as a side benefit, allows the peak switch voltage to be brought back inside the specified RBSOA region of the 6.5 kV IGBT modules.

1 Problem statement

The test bench sequence is: a pre-heating phase at low voltage with 3000 A DC through the load, followed by the application of $V_s = 3\,\mathrm{kV}$ to the series R_L–L load ($R_L = 0.8\,\Omega$, $L = 2\,\mathrm{mH}$, $\tau_L = L/R_L = 2.5\,\mathrm{ms}$) for 0.1 s. At the end of the pulse the solid-state switch (four paralleled 5SNA 1000G650300 6.5 kV HiPak IGBT modules [5]) interrupts the current, whose simulated peak is $I_0 \approx 3600\,\mathrm{A}$. By specification there is no free-wheeling diode across the load: the current must be forced to zero against a counter-voltage. In the baseline design this counter-voltage is provided by a MOV bank (EPCOS/TDK B80K680, 3 series × 5 parallel strings) clamping the switch voltage at $V_c \approx 5.5\,\mathrm{kV}$.

The service-life target of the bench is on the order of 10^6 or more interruption cycles. This document examines why the MOV bank cannot carry the cyclic energy for that many operations, and details the dump-resistor alternative.

2 Energy budget of the MOV clamp

2.1 Extinction transient

With the MOV clamping the *switch* voltage at V_c, the source remains in the loop during the extinction. The loop equation is

$$L\frac{di}{dt} = V_s - V_c - R_L\, i, \qquad i(0) = I_0, \tag{1}$$

whose solution is

$$i(t) = i_\infty + (I_0 - i_\infty)\, e^{-t/\tau_L}, \qquad i_\infty = \frac{V_s - V_c}{R_L}. \tag{2}$$

With the numbers above, $i_\infty = (3000 - 5500)/0.8 = -3125\,\mathrm{A}$ and the current reaches zero at

$$t_z = \tau_L \ln\frac{I_0 - i_\infty}{-i_\infty} = 2.5\,\mathrm{ms} \cdot \ln\frac{6725}{3125} \approx 1.9\,\mathrm{ms}. \tag{3}$$

The charge delivered through the MOV during the extinction is

$$Q = \int_0^{t_z} i\,dt = i_\infty t_z + (I_0 - i_\infty)\,\tau_L\Big(1 - e^{-t_z/\tau_L}\Big) \approx 3.0\,\mathrm{C}. \tag{4}$$

2.2 Energy balance

The energy absorbed by the MOV bank per operation is

$$E_{\mathrm{MOV}} = V_c\, Q \approx 5500 \times 3.01 \approx 16.6\,\mathrm{kJ}, \tag{5}$$

which exceeds the inductive energy $\frac{1}{2}LI_0^2 = 12.96\,\mathrm{kJ}$ because the source keeps feeding the loop while the clamp is active: the source delivers $E_s = V_s Q \approx 9.0\,\mathrm{kJ}$ during the transient, the load resistor dissipates $\approx 5.4\,\mathrm{kJ}$, and the MOV takes the remaining 16.6 kJ.

Distributed over the bank, each of the five parallel strings absorbs $\approx 3.3\,\mathrm{kJ}$ and each disc $\approx 1.1\,\mathrm{kJ}$ per operation, with a peak string current of 720 A.

2.3 Why this is incompatible with 10^6 operations

Three independent mechanisms limit the MOV in this duty [3]:

1. **Repetitive-pulse rating.** Varistor data sheets specify the maximum absorbed energy for a small number of pulses (typically 1 to 10^3, with a strong derating as the count grows). An absorption of the order of half the single-pulse class, repeated 10^6 times, is far outside any published derating curve.

2. **Cumulative degradation.** Every surge shifts the varistor characteristic: $V_{1\,\mathrm{mA}}$ decreases and the leakage current at the standing voltage grows. The leakage region has a negative temperature coefficient, so increasing leakage at 3 kV standing voltage leads toward thermal runaway.

3. **Parallel-string instability.** With a typical $\pm 10\%$ tolerance on $V_{1\,\mathrm{mA}}$ and the extremely steep V–I characteristic, the lowest string absorbs a disproportionate share of the energy, degrades faster, and its share grows further. Parallel MOV strings under heavy cyclic duty are unstable by construction unless the discs are individually matched, and even then the matching drifts.

Finally, the *average* power matters. At one operation per minute the bank dissipates $16.6\,\mathrm{kJ}/60\,\mathrm{s} \approx 280\,\mathrm{W}$, i.e. $\approx 18\,\mathrm{W}$ per disc continuously — close to or above the admissible average power of a block varistor, again pushing toward thermal runaway. The conclusion is that in the baseline design the MOV bank is a *consumable*, not a component.

3 Dump-resistor concept

3.1 Topology

The idea is borrowed from the energy-extraction systems used to protect superconducting magnet circuits [4]: at the interruption command the stored magnetic energy is diverted into a linear resistor with large thermal mass, whose electrical life is practically unlimited.

Figure 1 shows the arrangement. A crowbar branch — a series connection of the dump resistor R_D, a small commutation inductance L_c (partly the parasitic inductance of the resistor itself) and a crowbar device — is connected *across the load*. During normal operation the branch is blocked. At the interruption command the crowbar is armed, the main switch turns off, and the load current commutates into the dump branch, where it decays against the counter-voltage $R_D\, i$.

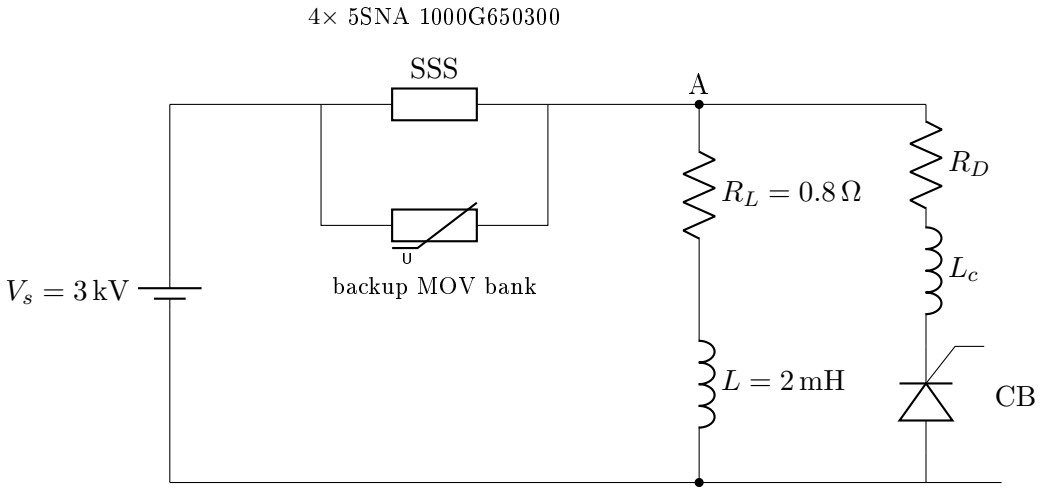


Figure 1: Dump-resistor energy extraction. The crowbar branch (R_D, L_c, crowbar device CB) is connected across the load; the MOV bank remains across the solid-state switch (SSS) as a backup clamp only. The crowbar device conducts upward (from the return rail into node A), i.e. it is reverse-biased by V_s during normal conduction.

3.2 Why across the load and not across the switch

If the dump branch were placed across the *switch*, the source would remain in the extinction loop exactly as in the MOV case of Section 2: the source would feed additional energy into the event and the resistor would have to absorb it. With the branch across the *load*, the turn-off of the main switch disconnects the source completely: the total energy dissipated in the extinction is exactly $\frac{1}{2}LI_0^2 = 12.96\,\mathrm{kJ}$, shared between R_L and R_D in proportion to their resistances. This placement is therefore strictly better on every count.

4 Circuit analysis

4.1 Decay transient

After commutation the loop is L–R_L–R_D (the source is disconnected):

$$i(t) = I_0\, e^{-t/\tau_d}, \qquad \tau_d = \frac{L}{R_L + R_D}. \tag{6}$$

4.2 Switch voltage

During the decay the main switch blocks

$$v_{\mathrm{sw}}(t) = V_s + R_D\, i(t) + v_{\mathrm{CB}}(t) \xrightarrow{t\to\infty} V_s, \tag{7}$$

so the peak switch voltage occurs at the first instant,

$$\hat{V}_{\mathrm{sw}} = V_s + R_D\, I_0, \tag{8}$$

and then *decays together with the current*, unlike the MOV clamp which holds the switch at V_c for the whole extinction. Equation (8) is the design equation for R_D: the resistor is chosen as the largest value compatible with the switch voltage budget.

4.3 Energy split

$$E_{\mathrm{tot}} = \tfrac{1}{2} L I_0^2, \qquad E_D = E_{\mathrm{tot}}\,\frac{R_D}{R_L + R_D}, \qquad E_{R_L} = E_{\mathrm{tot}}\,\frac{R_L}{R_L + R_D}. \tag{9}$$

Note that the load resistor already dissipates $V_s I_0 \approx 10.8\,\mathrm{MW}$ during the 0.1 s pulse, i.e. about 1 MJ per cycle: the few additional kilojoules of the extinction are negligible for it.

4.4 Worked comparison

Table 1 compares the MOV baseline with three values of R_D, including $R_D = 0$ (plain crowbar short, shown only as a reference: it corresponds to the excluded free-wheeling behaviour). The time to reach 1% of I_0 is $t_{1\%} = \tau_d \ln 100 = 4.6\,\tau_d$. Figure 2 plots the current decays.

Table 1: Interruption of $I_0 = 3600\,\mathrm{A}$: MOV clamp vs. dump resistor. E_{abs} is the energy in the absorber (MOV bank or R_D); E_{R_L} the extinction energy in the load resistor.

	$\hat{V}_{\mathrm{sw}}$	τ	current zero	E_{abs}	E_{R_L}
MOV clamp $V_c = 5.5\,\mathrm{kV}$	5.5 kV (held)	2.5 ms	1.9 ms (hard)	16.6 kJ	5.4 kJ
$R_D = 0.70\,\Omega$	5.52 kV (decaying)	1.33 ms	$t_{1\%} = 6.1$ ms	6.0 kJ	6.9 kJ
$R_D = 0.42\,\Omega$	4.51 kV (decaying)	1.64 ms	$t_{1\%} = 7.6$ ms	4.5 kJ	8.5 kJ
$R_D = 0$ (reference)	3.0 kV	2.5 ms	$t_{1\%} = 11.5$ ms	0	13.0 kJ

Two observations:

- With $R_D \approx 0.7\,\Omega$ the extinction speed is essentially that of the MOV clamp (the current is below 5% of I_0 after 4 ms), the peak switch voltage is the same 5.5 kV but only instantaneous, and the absorber energy per operation drops from 16.6 kJ in a degrading nonlinear component to 6.0 kJ in a linear metallic one.
- With $R_D \approx 0.42\,\Omega$ the peak switch voltage falls to 4.5 kV, i.e. back inside the region where the RBSOA of a 6.5 kV HiPak is actually specified [5, 7]. The dump resistor therefore solves the RBSOA overreach of the 5.5 kV MOV clamp at the same time as the life problem, at the cost of a moderately longer current tail.

4.5 Reaching an actual current zero

The exponential (6) never reaches zero mathematically; in practice the crowbar thyristor turns off when the current falls below its holding current I_H (fractions of an ampere), at $t \approx \tau_d \ln(I_0/I_H) \approx 15$ ms, which is a true, arc-free current zero.

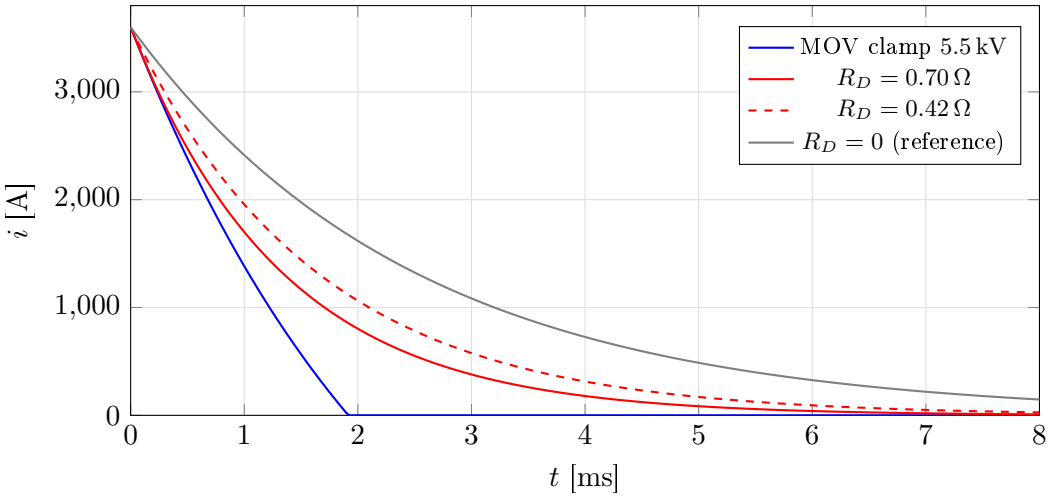


Figure 2: Current decay after switch turn-off. The MOV clamp reaches a hard zero in 1.9 ms; the dump resistor gives an exponential decay of comparable speed. With $R_D = 0.42\,\Omega$ the peak switch voltage stays at 4.5 kV.

If the application requires a hard zero on the millisecond scale, a small varistor of clamping voltage V_M can be placed in series with R_D. The loop equation becomes affine,

$$L\frac{di}{dt} = -(R_L + R_D)\, i - V_M, \tag{10}$$

with a finite extinction time

$$t_z = \tau_d \ln\!\left(1 + \frac{I_0(R_L + R_D)}{V_M}\right). \tag{11}$$

This option should be weighed carefully: the series varistor absorbs $E_M = V_M Q$ with Q of the order of 3 C, i.e. several kilojoules for V_M in the kilovolt range — it merely reintroduces the MOV duty at reduced scale. It is preferable to accept the exponential tail and the holding-current turn-off, unless the test specification explicitly demands otherwise.

5 Commutation into the dump branch

This is the one genuinely critical design detail of the scheme. The main switch turns off in about a microsecond; the load current must therefore be picked up by the crowbar branch at a rate that the crowbar device tolerates. Two implementations are proposed.

5.1 Option A: thyristor crowbar

A single phase-control thyristor, e.g. the 6.5 kV 5STP 26N6500 [6], comfortably provides the required ratings: reverse blocking of $V_s = 3$ kV with a factor-2 margin, surge capability far above 3600 A for milliseconds, and natural turn-off at the holding current. Points of attention:

Gating while reverse-biased. Before the main switch opens, the thyristor sits reverse-biased at $-V_s$ and cannot be fired. The gate must be driven with a pulse train (with back porch) starting shortly before the turn-off command, so that the device latches as soon as it becomes forward-biased. Gate current applied under reverse bias increases the reverse leakage and its dissipation; the arming lead time should therefore be kept short (e.g. about 100 µs), which is trivial since the interruption instant is commanded by the bench itself.

Turn-on di/dt. A phase-control thyristor tolerates a repetitive turn-on di/dt of the order of 100 A/µs to 300 A/µs (data-sheet value to be respected at 10^6 repetitions), whereas the raw commutation would demand several kA/µs. A series commutation inductance L_c limits the rate. During the transfer the backup MOV holds the switch at V_c, so the voltage driving the current into the branch is $V_c - V_s \approx 2.5\,\mathrm{kV}$:

$$\left.\frac{di_{\mathrm{CB}}}{dt}\right|_0 = \frac{V_c - V_s}{L_c}, \qquad L_c = \frac{2.5\,\mathrm{kV}}{200\,\mathrm{A/\mu s}} \approx 12.5\,\mu\mathrm{H}, \tag{12}$$

an inductance largely provided for free by a grid-type dump resistor and its connections. The transfer completes in some tens of microseconds; during this interval the MOV bank carries the difference current and absorbs

$$E_{\mathrm{MOV,comm}} \approx V_c \int (i_L - i_{\mathrm{CB}})\, dt \approx 200\,\mathrm{J} \text{ to } 350\,\mathrm{J} \tag{13}$$

per operation (the exact value depends on R_D and L_c; with $R_D = 0.42\,\Omega$, $L_c = 12.5\,\mu\mathrm{H}$ the transfer takes $\approx 28\,\mu\mathrm{s}$ and the MOV absorbs $\approx 230\,\mathrm{J}$). Spread over the bank this is 15 J to 25 J per disc per operation, i.e. about 2% of the single-pulse class: a level that the repetitive derating curves of block varistors may admit at 10^6 pulses, *to be verified on the data-sheet derating chart* [3]. A saturable commutation reactor (high unsaturated inductance for the first microseconds, then saturating) reduces this residual MOV duty further, a standard technique in pulse-power circuits.

Re-applied voltage. After extinction the branch sees no voltage (the switch is open); reverse voltage returns only at the next closing of the main switch, many seconds later, so the circuit-commutated recovery time t_q is irrelevant. An RC snubber across the thyristor handles the re-applied dv/dt at switch closing.

5.2 Option B: pre-triggered IGBT crowbar

The crowbar can instead be an IGBT (or two paralleled modules of the same 5SNA 1000G650300 family) in series with a blocking diode oriented to conduct from the return rail into node A. During normal operation the diode blocks V_s; the crowbar IGBT can simply be held *on* for the whole armed period. At main-switch turn-off the branch picks up the current with no firing delay and no di/dt limitation other than the loop stray inductance: the residual MOV duty of Option A essentially disappears.

Ratings: with two paralleled modules each carries 1800 A peak, inside the 1 ms pulsed collector current rating $I_{CM} = 2000\,\mathrm{A}$, and decaying. The conducted charge is $Q = I_0\tau_d \approx 6\,\mathrm{C}$, so the conduction energy per operation is $\sim V_{CE(\mathrm{sat})}Q \approx 15\,\mathrm{J}$ — thermally negligible, with junction temperature excursions below 1 K per shot and therefore no power-cycling concern. The cost is one more gate driver, one blocking diode under 3 kV DC (same cosmic-ray considerations as the main switch [8]), and two more modules.

Option A is simpler and more rugged; Option B is cleaner electrically. Both are compatible with the life target.

6 Dump resistor design

Energy and temperature per pulse. With $E_D = 4.5\,\mathrm{kJ}$ to 6.0 kJ per operation, an adiabatic element of mass m and specific heat c rises by $\Delta T = E_D/(mc)$ per pulse. A stainless-steel or FeCrAl grid element of 20 kg ($c \approx 480\,\mathrm{J/(kg\,K)}$) rises by only about 0.6 K per pulse: thermal fatigue is a non-issue.

Average power. The average dissipation is $\bar{P} = E_D/T_{cyc}$: 100 W at one operation per minute, 600 W at one per ten seconds — trivially handled by natural or light forced convection. The cycle period of the bench should be confirmed, as it sets this number (and it is the same number that would drive the MOV bank into thermal runaway).

Technology. Grid or ribbon resistors of the braking-resistor class are the natural choice: metallic conduction, no drift mechanism, low temperature coefficient, and a parasitic inductance of a few microhenries that is actually *useful* in Option A (it forms part of L_c). For Option B a lower-inductance construction (Ayrton–Perry or bifilar winding) is preferable, with the residual $L_R\, di/dt$ overshoot at commutation covered by the module snubbers and the backup MOV.

Life. The cumulative energy over 10^6 operations is ~ 6 GJ — meaningless for a metallic element sized as above, whereas the equivalent cumulative charge of ~ 3 MC through the MOV bank is far beyond any varistor capability. This asymmetry is the core argument for the scheme.

7 Residual role of the MOV bank

The MOV bank is retained across the switch, but demoted from per-cycle absorber to *backup clamp*. Its duties become:

1. clamping the first microseconds of each commutation (Option A only, 200 J to 350 J per operation, to be verified against the repetitive derating curve, or reduced with a saturable reactor);
2. absorbing the full 16.6 kJ event if the crowbar fails to fire. The existing 3s × 5p B80K680 bank is already sized for this as a single event; the data-sheet surge count at this level (tens of pulses) becomes an acceptable *fault budget* rather than a life-limiting consumable.

Practical measures: count crowbar misfires (a misfire is directly observable from the switch-voltage waveform), periodically verify $V_{1\,\mathrm{mA}}$ of the strings, and alarm on drift. A crowbar device failed short is also benign and observable: it appears as a resistive divider across the load at the next switch closing.

8 Conclusions

- The baseline MOV clamp absorbs ≈ 16.6 kJ per operation — more than the stored inductive energy, because the source keeps feeding the clamped loop. No varistor technology sustains this duty for 10^6 cycles; drift, parallel-string imbalance and average-power thermal runaway all point the same way.
- A dump resistor switched across the load by a crowbar extracts the energy with the source disconnected: only $\frac{1}{2}LI_0^2$ is dissipated, split between load and dump resistors, in components with practically unlimited electrical life.
- Extinction speed is preserved: with $R_D = 0.7\,\Omega$ the decay is as fast as the MOV clamp; with $R_D = 0.42\,\Omega$ the peak switch voltage drops to 4.5 kV, restoring margin to the specified RBSOA of the 6.5 kV modules.
- The critical implementation detail is the commutation di/dt: solved with a 10 µH to 15 µH commutation inductance (thyristor crowbar, Option A) or eliminated with a pre-triggered IGBT + blocking diode branch (Option B).
- The MOV bank remains as a backup clamp with a fault budget of tens of full events, plus (Option A) a small per-cycle commutation duty to be checked against the repetitive derating curves.

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